

Queensland Community Pharmacy Chronic Conditions Management Pilot

Chronic Obstructive Pulmonary Disease (COPD) Monitoring Program

Pilot Clinical Protocol – V2

Eligibility for the Program

Eligibility for participation in the Chronic Obstructive Pulmonary Disease (COPD) Program (the Program) must be assessed at each consultation, as eligibility may change due to changes in health status or demographic factors. i.e. patients who were previously eligible may become ineligible and patients who were ineligible may become eligible.

Refer the patient to their usual healthcare provider (or emergency services, if required) with a comprehensive clinical handover if the patient is, or becomes, ineligible for management under the Program. Ineligibility for treatment as part of a Pilot service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Manage patients with symptoms and features of mild COPD and asthma (including those with a diagnosis of co-existing asthma and COPD) under the clinical protocol for the Improved Asthma Symptom Control Program. Confirm that the patient meets the eligibility criteria of the COPD Monitoring Program before commencing treatment.



Patients who are ineligible for the Program:

Patients who:

- are aged <35 years
- are planning a pregnancy or are pregnant
- are experiencing respiratory distress (e.g., unable to complete a whole sentence in one breath)
- during/after exercise experience chest tightness or discomfort (without other symptoms of COPD) - consider possibility of a cardiovascular condition/event
- have clinical features, symptoms or findings suggestive of a respiratory condition other than COPD that is not suitable for management under the Program

- are currently receiving, or have received within the past 12 months, specialist treatment for COPD or another respiratory condition, unless they have a written referral from their treating specialist or general practitioner to participate in the Program
- are newly diagnosed with or develop new signs or symptoms of a mental health condition, or have an existing poorly controlled mental health condition (for example but not limited to depression, anxiety disorder)
- have contraindications to spirometry (that will not resolve with time)
- have spirometry results that:
 - are difficult to interpret or are unexpected for COPD
 - indicate alarm symptoms or findings
 - do not demonstrate persistent airflow limitation
 - show abnormal restrictive or mixed ventilatory patterns.
- have a forced expiratory volume in 1 second (**FEV₁**) of **<60% predicted**
- have COPD and:
 - have ever been **hospitalised** for their COPD
 - have ever had an exacerbation of their COPD requiring oral corticosteroids (OCS) **and** antibiotics
 - have had ≥ 2 COPD exacerbations (that required systemic corticosteroids **or** antibiotics to treat) in the previous 12 months.
- have complex cardiovascular disease:
 - severe (Grade 3) hypertension (≥ 180 mmHg systolic and/or ≥ 110 mmHg diastolic)
 - congenital heart disease
 - rheumatic heart disease
 - heart failure
 - arrhythmias
 - atrial fibrillation
- have other conditions including peripheral arterial disease, heart block, pericarditis, valvular disease, pulmonary hypertension, angina, cardiomyopathy and cardiomegaly, aortic aneurysm, ischaemic heart disease, acute coronary syndrome, stroke or other cerebrovascular disease, deep vein thrombosis or pulmonary embolism.
- have any alarm signs and symptoms including haemoptysis, chest pain, unexplained weight loss or gain, night sweats, fever, signs of bronchiectasis, recurrent pneumonia, orthopnoea and oedema (prompt medical review is required)
- have any symptoms or features that suggest that COPD has progressed despite adequate management
- have type 1 diabetes
- have poorly controlled type 2 diabetes
- patients whose symptoms do not improve after one month of therapy or worsen following commencement of Step 3 long-acting bronchodilator dual therapy (long-acting muscarinic antagonist (LAMA) and long-acting beta agonist (LABA)), requiring initiation of triple inhaler therapy (addition of an inhaled corticosteroid (ICS)).



Treat (if clinically indicated) and concurrently refer

Provide treatment (if clinically indicated) and concurrently refer the patient to an appropriate healthcare provider for further review if they:

- have well controlled type 2 diabetes
- have a stable and well controlled mental health condition (e.g., depression, anxiety disorder)
- show signs and symptoms of cognitive impairment or are diagnosed with dementia or another neurocognitive condition
- have a suspected bacterial infection that requires antibiotic therapy
 - patient may continue to be managed in the Program following concurrent referral to a medical practitioner for antibiotic therapy
 - where appropriate, support the patient to manage their acute exacerbation according to their COPD action plan.
- have been identified during case finding as being aged <45 years and have not been investigated for alpha1-antitrypsin deficiency.

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When applying the information contained within this clinical protocol, pharmacists should exercise professional discretion and judgement. The protocol supports, but does not replace, the pharmacist's responsibility to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their caregiver.

How to use this document

The purpose of the Program is to provide an accessible, community-based healthcare service to identify and improve outcomes for patients with symptoms consistent with mild COPD.

This clinical protocol details how patients in the Program should be managed, including referral to other health services and medical practitioners, pharmacological and non-pharmacological interventions, and protocol-based/structured prescribing of medicines for the management of COPD.

Overview of the COPD Monitoring Program

There are two entry points for patients for enrolment in the Program:

Entry point 1:

- Patients **without** a previous diagnosis of COPD, who do not meet any ineligibility criteria, may be enrolled in the Program and **commence management** (with non-pharmacological measures and pharmacotherapy, if indicated) **concurrent to referral** to an appropriate healthcare provider for further review and collaborative care.

Entry point 2:

- Patients **with** an existing diagnosis of COPD, who do not meet any ineligibility criteria, may be enrolled in the Program and **commence ongoing monitoring and management** (with non-pharmacological measures and pharmacotherapy, if indicated). **An update must be provided** to the patient's usual healthcare provider following each occasion of care.

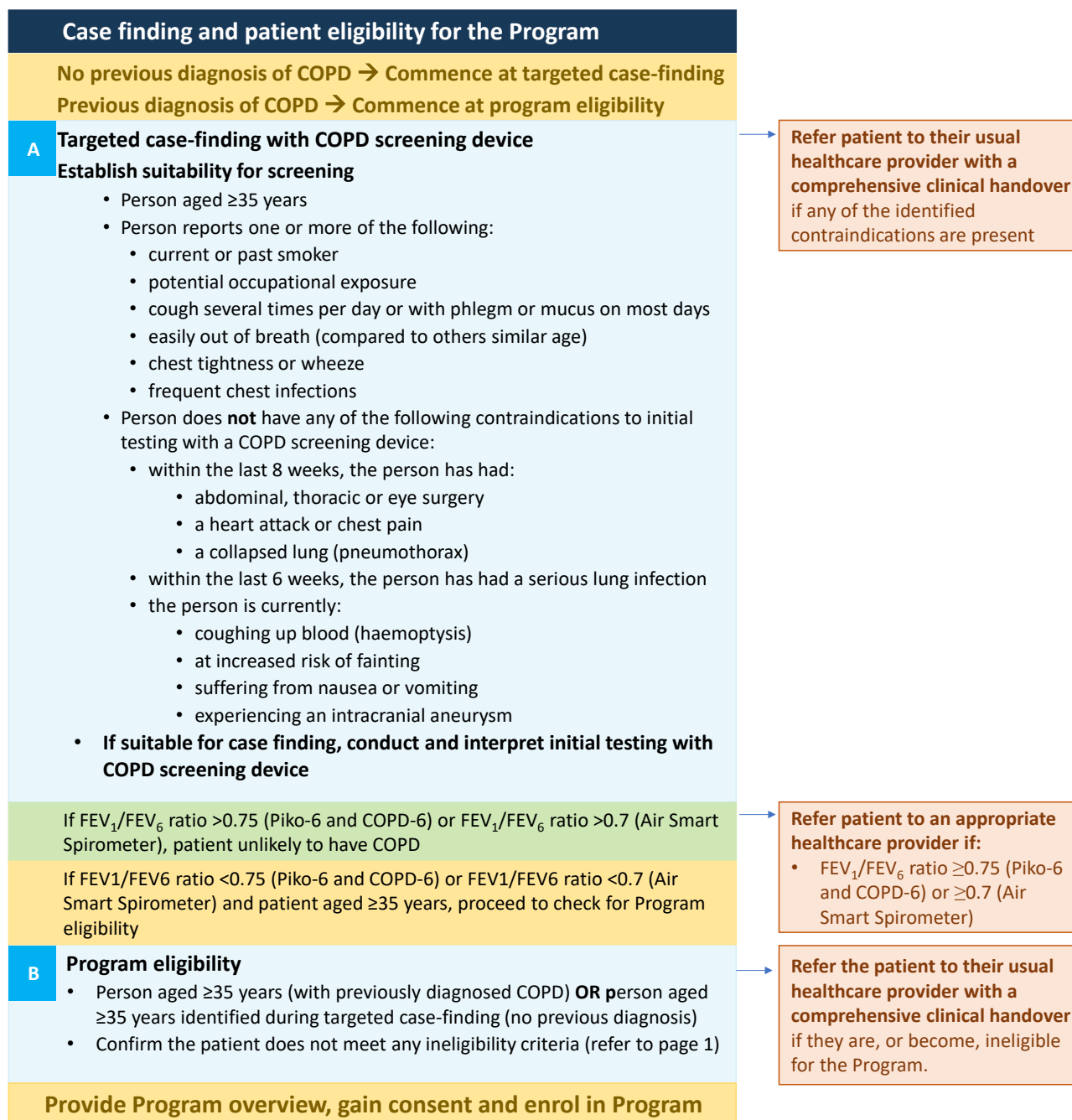
[Figure 1](#) provides an overview of the Program, which includes:

- **case finding and program eligibility** (and referral if indicated)
- **structured assessment**
- **management of mild COPD** (defined as FEV₁ between 60% and 80%) including **development and use of a COPD Action Plan**
- **ongoing monitoring** of disease progression with referral to appropriate healthcare providers for collaborative management as required.

When a patient enters the Program:

- Provide an overview of the Program (see [Figure 1](#)), including the aims, expected outcomes and what the Program may involve (which will vary between patients), including:
 - timeframes for management of the condition, such as the number of appointments and testing required, and the costs involved
 - how the patient's medicines may be managed and how medicine costs may differ when prescribed by a pharmacist
 - other interventions that may be recommended as part of the Program e.g., smoking cessation and weight management, pulmonary rehabilitation and lung cancer screening
 - that the patient may leave the Program, including by opting out or becoming ineligible, at any time, and may be referred to an appropriate healthcare provider.
- Document informed consent from the patient for participation, as per the [Pilot Handbook](#).

Figure 1: Overview of the COPD monitoring program



See over page

Patient history and assessment	
C	Take a detailed patient history • Patient characteristics • Medical history • Lifestyle/ social history • Review previous test results (if available) If the person has no previous diagnosis of COPD – consider contraindications to spirometry
D	Assess severity and impact of COPD symptoms (including Chronic Airways Assessment Test)
E	Conduct physical examination
F	Conduct (or refer for) spirometry testing (if applicable)
G	Interpret spirometry (if applicable) • Persistent airflow limitation ($FEV_1/FVC \leq 0.7$, with an FEV_1 of $<80\%$ of the predicted value (post-bronchodilator)) • Mild COPD: $FEV_1 = 60$ to 79% of predicted lung function • Moderate COPD: $FEV_1 = 40$ to 59% of predicted lung function • Severe COPD: $FEV_1 = <40\%$ of predicted lung function
Commence collaborative care	
Develop and implement management plan	
H	Non-pharmacological measures • Manage risk factors • Avoidance of exposure to risk factors • Smoking cessation • Immunisation • Optimising function • Physical activity • Pulmonary rehabilitation • Diet and nutrition
I	Pharmacotherapy • Stepwise pharmacological management of stable mild COPD • Steps 1–3 as per the <i>Therapeutic Guidelines</i> • Managing exacerbations (self-management) • Initial and secondary pharmacotherapy: inhaled bronchodilator and if appropriate prednis(ol)one as per the <i>Therapeutic Guidelines</i>
J	Confirm management plan is appropriate
K	Communicate agreed management plan
L	Collaborative care
Ongoing management and monitoring	
M	Clinical review • 3 months after commencement of the Program or after initiation of/change to pharmacotherapy • 6 months for patients with stable pharmacotherapy and COPD stable
The patient's eligibility for the Program may change at any point. Refer the patient to their usual healthcare provider with a comprehensive clinical handover should the patient become ineligible for management under the Program or choose to exit the Program.	

Refer to an appropriate healthcare provider if spirometry results:

- Are inconsistent with COPD (or difficult to interpret)
- Indicate a respiratory condition other than, or coexisting with COPD that is not suitable for management under the Program
- Indicate alarm symptoms or findings or show abnormal restrictive or mixed ventilatory patterns
- Indicate moderate or severe COPD

Refer patient to their usual healthcare provider with a comprehensive clinical handover if symptoms are not adequately managed with pharmacotherapy at Step 3 (indicating possible progression to moderate-severe COPD or an alternative cause for symptoms).

Key points

- COPD is a progressive obstructive respiratory disease characterised by persistent airflow limitation and respiratory symptoms due to small airway abnormalities and/or alveolar destruction (emphysema) ⁽¹⁻⁴⁾.
- COPD is caused by an abnormal inflammatory response in the lungs generally as a result of exposure to noxious particles or gases from tobacco smoke (either directly or passively) or other environmental exposures to air pollution or occupational dust ⁽¹⁻⁵⁾.
- COPD is preventable and treatable, but not fully reversible. Management is stepwise and aims to:
 - relieve and control symptoms
 - treat exacerbations and prevent future exacerbations
 - improve exercise tolerance
 - improve health-related quality of life
 - delay/slow the progression of the disease ⁽²⁻⁴⁾.
- Patients with COPD may also experience exacerbations/flare ups (acute worsening of symptoms that extends beyond expected day-to-day symptom variation) that increase in frequency as the disease progresses and require additional therapy ^(1, 6). Exacerbations may be:
 - mild (treated with short-acting bronchodilators and usually self-managed)
 - moderate (requiring systemic corticosteroid treatment and/or antibiotic therapy)
 - severe (requiring hospitalisation) ⁽¹⁾.
- COPD disproportionately affects Aboriginal and Torres Strait Islander peoples, who experience nearly twice the prevalence and three times the mortality rate compared to non-Indigenous Australians. People in rural and remote areas also face higher rates of COPD, with poorer outcomes linked to inequitable access to health services ⁽⁷⁾.

Case finding and Program eligibility

A: Targeted case-finding

- Conduct targeted case-finding and, if suitable, conduct and interpret initial testing with a COPD screening device for patients **without an existing diagnosis of COPD** (refer to [Figure 1](#)).

Note: this is for screening purposes only and does not replace spirometry to determine diagnosis.

- COPD screening using COPD screening devices should be guided by the [Lung Foundation Australia Position paper: Use of COPD screening devices for targeted COPD case finding in community settings](#) ⁽⁸⁾.
- [The Community Pharmacy COPD case finding results form](#) can be used to record the results of case finding ⁽⁹⁾.

B: Program eligibility

- Confirm eligibility to participate in the Program (see [Eligibility](#)).
- Consider program eligibility at every consultation as eligibility may change at any time.

- Refer the patient to their usual healthcare provider with a comprehensive clinical handover if the patient is, or becomes, ineligible for management under the Program.
- Consider smoking cessation for all current smokers. Refer to the [Smoking Cessation Clinical Practice Guideline](#) ⁽¹⁰⁾.

Patient history and assessment

C: Patient history

Obtain sufficient information to assess the patient's condition, and the safety and appropriateness of any recommendations and medicines for the patient.

Consider:

- age
- ethnic or cultural background, including Aboriginal and/or Torres Strait Islander status
- lactation status
- previous and current medical conditions, including systemic and/or chronic conditions and those that may be contraindications to spirometry (refer to [F. Conduct \(or refer for\) spirometry testing](#))
- screening for conditions linked to worsening or exacerbation of COPD including but not limited to undiagnosed cardiovascular disease, sleep apnoea, gastro-oesophageal reflux disease (GORD), cognitive impairment, diabetes and mental health conditions especially anxiety and depression ⁽¹⁾.
- history of respiratory tract infections in the previous 12 months
- presence of COPD risk factors (refer to [Appendix 1](#)) including environmental and occupational exposures, smoking history and current status (consider tobacco, cannabis, vaping, passive smoking), and family history of COPD
- previous diagnosis of COPD:
 - date of diagnosis, last medical review for COPD, most recent spirometry testing
 - history of COPD exacerbations in the previous 12 months
 - history of COPD exacerbations requiring systemic corticosteroids and/or antibiotics
 - hospitalisations for COPD
 - current (or most recently prescribed) treatment and most recent COPD Action Plan
 - current or previous pulmonary rehabilitation.
 - presence, pattern and severity of all respiratory symptoms (refer to [Table 1](#)):
 - frequency and time of day
 - impact on daily activities
 - triggers and precipitating factors
 - relieving factors e.g., short acting beta agonist (SABA) or short acting muscarinic (SAMA) inhaler use
- alarm signs and symptoms that require further investigation e.g., haemoptysis, chest pain, unexplained weight loss or gain, night sweats, fever, signs of bronchiectasis, recurrent pneumonia, orthopnoea or oedema (see [Eligibility](#))
- all current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements and over-the-counter medicines)

- drug and non-drug allergies, and adverse drug reactions
- diet/nutrition status and weight
- levels of physical activity
- recreational or illicit drug and alcohol use
- vaccination status (particularly pneumococcal, influenza, respiratory syncytial virus (RSV), COVID-19, pertussis and herpes zoster) ⁽¹⁾
- specialist involvement in care for any respiratory condition
- co-morbidities or factors that may be associated with COPD that may require additional management to improve COPD and general health outcomes (see [GOLD 2026 Report Pocket Guide](#) for approaches to detection and review of frequent co-morbidities) ^(1, 4):
 - periodontitis and oral disease
 - osteoporosis
 - GORD
 - vitamin D levels
 - anaemia
 - polycythaemia
 - elevated white cell count
 - falls risk
 - sleep apnoea and other sleep related disorders
 - dysphagia
 - use of alcohol and sedative medication
 - cognitive impairment.

Table 1. Symptoms of COPD ^(1, 2, 4)

- Persistent breathlessness (dyspnoea) that is unexplained or inappropriate for level of exertion, as per the [Modified Medical Research Council Dyspnoea Scale](#) (most characteristic symptom of COPD).
- Chronic cough (several times most days for two months), intermittent, or unusual.
- Frequent or unusual sputum production (cough up phlegm or mucus most days).
- Recurrent wheeze or chest tightness.
- Fatigue and reduced exercise tolerance.
- Recurrent lower respiratory tract infections.

NB: Symptoms may be subtle and not noticeable to the patient until the disease has significantly progressed, and lung damage has occurred.

Co-existing asthma and COPD

COPD and asthma may co-exist, and many adult patients show features of both conditions ⁽¹¹⁾. Patients with a history of asthma are at increased risk of developing COPD ⁽¹⁾ and patients with COPD often report a history of asthma ^(1, 2).

A clear distinction between COPD and asthma can be difficult when features of both conditions are present, and further investigations may be required ⁽¹¹⁾. Clinical features that can aid in differentiation between asthma and COPD, or indicate possible co-existing asthma and COPD, are shown in [Appendix 2](#).

Manage patients with symptoms and features of mild COPD and asthma (including those with a diagnosis of co-existing asthma and COPD) under the clinical protocol for the Improved Asthma Symptom Control Program. Confirm that the patient meets the eligibility criteria for the COPD Monitoring Program and the Asthma Symptom Control Program before commencing treatment.



Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

D: Assess COPD severity and impact of symptoms

[Appendix 3](#) details the classification of stable COPD severity (i.e., preferably not during or within 4 weeks of an exacerbation) as per the current online version of the [Therapeutic Guidelines: Stable COPD](#) ⁽²⁾ and the [COPD-X Plan: Assessing the severity of COPD](#) ⁽⁴⁾.

Chronic Airways Assessment Test (CAAT)

The self-administered online [Chronic Airways Assessment Test \(CAAT\)](#) ⁽¹³⁾ should be used (in combination with patient reported symptoms and recent spirometry results) to assess disease severity and monitor disease progression.

The total CAAT score provides both clinicians and patients with a broad clinical picture of the impacts of COPD on the individual's life:

- very high impact = score of 31-40
- high impact = 21-30
- moderate impact = 10-20
- low impact = <10 ⁽⁴⁾.

The minimal clinically important difference (MCID) is a score of 2 ⁽⁴⁾.

If the patient reports symptoms consistent with moderate or severe COPD (refer to [Appendix 3](#)), or if symptoms are having a high or very high impact on the individual's life (determined through the CAAT score - see [CAAT user guide](#) for further information), they should be referred to a medical practitioner for management. **Do not proceed to physical examination.**

Modified Medical Research Council (mMRC) Dyspnoea Scale

The Modified Medical Research Council (mMRC) Dyspnoea Scale can be used to assess COPD symptoms. The mMRC assigns a grade based on the degree of functional disability due to breathlessness. For example, only breathless with strenuous exercise or walking slower because of breathlessness ⁽¹⁴⁾. See [MRC Dyspnoea Scale](#) for further information.

E: Physical examination

Perform a physical examination including chest auscultation. However, physical examination is not considered sensitive enough to detect/confirm mild COPD. Chest auscultation may be normal and wheeze is not an indicator of severity ⁽⁴⁾.

F: Conduct (or refer for) spirometry testing

Clinical signs and use of screening devices alone are not sufficient to confirm the presence, severity and reversibility of airflow limitation for symptoms consistent with mild COPD. Recent spirometry results including acute bronchodilator reversibility test results (within the previous 12 months) are required.

If spirometry testing has been conducted within the previous 12 months, obtain the results from the patient (if available) to avoid duplication.

Pharmacists may conduct spirometry in the community pharmacy setting in accordance with National Asthma Council Australia [Spirometry Handbook for Primary Care](#) ⁽¹⁵⁾ (the Spirometry Handbook), if they have been appropriately trained and have the appropriate equipment detailed in the Spirometry Handbook. Alternatively, refer to an appropriate healthcare provider that performs lung function testing.

If unable to conduct spirometry in the community pharmacy setting and access to an external healthcare provider performing spirometry is delayed, manage the patient under the Program until appropriate spirometry can be undertaken and the results reviewed.

Do not delay COPD treatment if clinically indicated while awaiting spirometry.

Conduct all spirometry in accordance with the Spirometry Handbook ⁽¹⁵⁾. Spirometry is generally safe, however, as it involves maximal effort, patients may experience effects such as bronchospasm, transient breathlessness, cough, syncope or chest pain ⁽¹⁵⁾. Discontinue spirometry if the patient experiences pain during testing ⁽⁸⁾.

The [Spirometry Handbook](#) outlines situations where spirometry is contraindicated or not recommended at a specific point of time. This can be because of risk to the patient or an increased likelihood of inaccurate or unrepeatable results ⁽⁸⁾.

If there is any concern or doubt about the safety or suitability of performing spirometry, refer the patient to an appropriate healthcare provide for management ⁽⁸⁾.

Resources for conducting spirometry

- National Asthma Council Australia [Spirometry Handbook for Primary Care](#) ⁽¹⁵⁾
- [Therapeutic Guidelines: Respiratory \(Pulmonary Function Testing – Spirometry\)](#) ⁽¹⁶⁾
- American Thoracic Society and European Respiratory Society Technical Statement: [Standardization of Spirometry 2019 Update](#) ⁽¹⁷⁾
- Patient fact sheet: Getting ready for spirometry fact sheet [Spirometry fact sheet](#) ⁽¹⁵⁾
- [Chronic Obstructive Pulmonary Disease \(COPD\) Hub | Australian Commission on Safety and Quality in Health Care](#) ⁽¹⁸⁾.

G: Interpret spirometry

Interpret spirometry testing in accordance with the latest online version of [Therapeutic Guidelines: Spirometry](#) ⁽¹⁶⁾, the [COPD-X Plan: Spirometry](#) ⁽⁴⁾, the National Asthma Council Australia [Spirometry Handbook for Primary Care](#) ⁽¹⁵⁾. See [Appendix 4](#) for more information on spirometry results that correlate with a diagnosis of COPD and key considerations.

Develop and implement a management plan

Optimal management of COPD involves a multidisciplinary and collaborative team-based approach and includes:

- identification of management goals collaboratively with the patient
- avoidance of triggers and minimisation of risk factors
- education and counselling to enable self-management
- non-pharmacological measures and pharmacotherapy
- management of exacerbations
- regular symptom monitoring and review ⁽²⁾.

COPD Action Plan and lifestyle prescription

Each patient within the Program should have the following developed, consistent with the current online version of the *Therapeutic Guidelines* and the *COPD-X Plan*:

- an individualised COPD Action Plan (or their existing plan updated), using a validated template, such as the [Lung Foundation Australia - COPD Action Plan](#) ⁽¹⁹⁾
- an accompanying individualised lifestyle prescription detailing non-pharmacological measures See [Appendix 5: Program lifestyle prescription to accompany COPD Action Plan.](#)

H: Non-pharmacological measures

Provide all patients with information about non-pharmacological measures, regardless of whether pharmacotherapy is prescribed.

Non-pharmacological measures alone may be sufficient for initial management of patients with stable COPD and minimal symptoms (e.g., a CAAT score < 10) ⁽²⁾.

Reducing risk factors

Avoidance of exposure to risk factors

Advise patients about the importance of avoiding risk factors, the most critical of which is smoking (including passive smoking) ^(2, 4). Avoidance of other environmental risk factors is also recommended, including occupational dust, fumes and gases, indoor and outdoor air pollutants ⁽⁴⁾.

Smoking and vaping cessation

Smoking and/or vaping cessation is the single most important and beneficial measure to prevent and/or limit lung damage, reduce exacerbations and reduce mortality for patients with COPD ^(2, 4).

Detailed guidance for pharmacists supporting smoking cessation is contained in the [Smoking Cessation Clinical Practice Guideline](#) and supporting vaping cessation is contained in the [E-cigarette and Vaping Cessation Guide](#) ⁽²⁰⁾.

Immunisation

Consider the immunisation status of patients with COPD, including for [pneumococcal](#) disease, [RSV](#), [COVID-19](#), [influenza](#), [pertussis](#) and [herpes zoster](#). Provide advice regarding the appropriateness based on the individual clinical circumstances ^(1, 2, 4). Refer to the Australian Immunisation Handbook for recommendations ⁽²¹⁾.

All vaccinations should be administered in accordance with the Australian Immunisation Handbook, individual vaccine information provided by the TGA, current Australian Technical Advisory Group on Immunisation (ATAGI) advice and the current version of the *Medicines and Poisons (Medicines) Regulation 2021*, available on the [Queensland Health website](#) ⁽²²⁾.

Optimising function

Physical activity

Many patients with COPD have a sedentary lifestyle and find physical activity difficult due to exertional dyspnoea and fatigue. However, there is good evidence that physical activity can reduce the risk of hospitalisations and improve quality of life ⁽⁴⁾.

Encourage patients with COPD to undertake physical activity in line with recommendations for their age group in the [Physical activity and exercise guidelines for all Australians](#) ⁽²³⁾.

- Not all patients will be able to meet the recommendations, however they should be advised that doing any amount of physical activity is better than none and encouraged to gradually build up to the recommendations ⁽²³⁾.

Encourage patients experiencing difficulty with physical activity to see a physiotherapist or exercise physiologist.

Referral for pulmonary rehabilitation

Pulmonary rehabilitation improves exercise capacity, mental health and quality of life regardless of COPD severity and can reduce the frequency of hospitalisations for exacerbations ^(2, 6).

Encourage patients with COPD to participate in a pulmonary rehabilitation program when symptomatic and after an exacerbation.

Information about pulmonary rehabilitation programs (and other services including psychosocial support) available throughout Queensland, including how referrals can be made, can be found on the [Lung Foundation Australia - Find a Service](#) website ⁽²⁴⁾.

Diet and nutrition

Malnutrition, obesity and poor nutritional status, generally resulting from suboptimal eating habits, sedentary lifestyle, smoking and corticosteroid use is common in patients with COPD, and has been shown to contribute to morbidity and mortality ^(1, 2, 4).

Advice should focus on a balanced diet with foods from each of the 5 food groups in appropriate portions to maintain a healthy weight, in accordance with the [Australian Dietary Guidelines](#) ⁽⁴⁾.

Weight management

Encourage underweight and overweight patients to see a dietitian for tailored nutritional advice.

Detailed guidance for pharmacists supporting weight management is contained in the [Management of Overweight and Obesity Clinical Practice Guideline](#) ⁽¹⁰⁾.

Referral for lung cancer screening

Refer eligible patients for screening under the National Lung Cancer Screening Program: [National Lung Cancer Screening Program information for healthcare providers](#) ⁽²⁵⁾.

I: Pharmacotherapy

As COPD progresses, management using non-pharmacological measures alone may become inadequate and pharmacotherapy may be required to relieve symptoms, reduce the frequency of exacerbations and improve overall quality of life ⁽²⁾.

Where pharmacotherapy is prescribed, the patient's prescription(s) (including repeats) should provide enough medicine for the period until the patient's next scheduled review.

Patients entering the program with an existing medicine regimen

Ask for the patient's most recent COPD Action Plan (if they have one) and consider whether the most recently prescribed pharmacotherapy is:

- appropriate for the patient's symptoms and medical history
- optimised for the patient, including inhaler technique and adherence.

When modifying pharmacotherapy within the Program that has been prescribed by another healthcare provider, attempt to make changes in collaboration with the original prescriber or the patient's usual healthcare provider (whichever is most appropriate).

Pharmacists may modify or deprescribe pharmacotherapy, particularly if:

- the patient's response is inadequate (within appropriate clinical timeframes)
- the patient is experiencing intolerable/unmanageable adverse effects
- patient is unable to use the prescribed device correctly after appropriate training
- modification is clinically warranted, and it is safe to do so.

When considering modification or adjustment to medicines prescribed, consider if the medicine is **also** used for the treatment of another condition (e.g., maintenance oral corticosteroids for another condition).

Stepwise pharmacotherapy for stable COPD

Pharmacotherapy for stable mild COPD within the Program must be in accordance with the current online version of the [Therapeutic Guidelines: Respiratory \(Stable chronic obstructive pulmonary disease \(COPD\) – Stepwise management of stable COPD\)](#) ⁽²⁾, specifically:

- [Step 1: Short-acting bronchodilator therapy](#)
- [Step 2: Long-acting bronchodilator monotherapy](#)
- [Step 3: Long-acting bronchodilator dual therapy.](#)

Combination products can be prescribed if appropriate, however they must only include medicines able to be prescribed under the Program.

Key considerations

- The entry point to the stepwise pathway should be individualised to the patient's previous medical and medication history. Patients with a new diagnosis of COPD or managing COPD with non-pharmacological measures should generally initiate pharmacotherapy at Step 1.
- When commencing any medicine, patients should be advised to return if they experience adverse effects, an increase in symptom frequency or type, more severe symptoms or an exacerbation.

Refer patients initiated on Step 3 Long-acting bronchodilator dual therapy (i.e. LAMA + LABA) where the condition has not stabilised or has worsened after 1 month to an appropriate healthcare provider for ongoing management.

Inhaler considerations

- The choice of medicine and device should consider patient preference and their ability to use the device correctly with appropriate training. Consider factors such as the inspiratory effort required, patient age, inspiratory effort capability, developmental stage, cognition and dexterity, and the environmental impact of the device (when in use and when disposed of) ⁽¹¹⁾.
- Recommend use of a spacer for people using pressured metered dose inhalers (with a face mask if required) to reduce issues with coordination, maximise lung deposition and reduce adverse effects of some medicines ⁽¹¹⁾.
- Minimise the number of inhalers prescribed, and the different types of inhaler devices, where possible to avoid confusion. If the use of more than one inhaler device type is unavoidable, provide clear instructions for correct device use and handling, and any differences (i.e. inspiratory effort required, shaking, storage, cleaning, disposal) ⁽¹¹⁾. Demonstrate the correct use of the patient's prescribed device (which may differ between brands) using a placebo inhaler. Provide an inhaler-specific checklist for the patient to follow ⁽¹¹⁾.
- Review inhaler technique and adherence at every opportunity, even if the same inhaler or inhaler device has been used long-term ^(11, 26).
- Further resources for consumers and health professionals are available in:
 - the National Asthma Council of Australia information paper for health professionals – [Inhaler technique for people with asthma or COPD](#) ⁽²⁶⁾
 - Lung Foundation inhaler technique checklist - [Inhaler technique: Device specific checklists - Lung Foundation Australia](#) ⁽²⁷⁾.

Management and treatment of COPD exacerbations

Advise all patients on how to recognise an exacerbation, including the signs and symptoms that indicate the need for emergency medical care, and to commence treatment at the onset of symptoms. This should be included in the patient's COPD Action Plan.

The [GOLD Report 2026](#) (Chapter 4: Management of exacerbations) has information to support pharmacists in assessing the severity of an exacerbation and differentiating conditions that may mimic or worsen exacerbation-like symptoms i.e., heart failure, pulmonary embolism ⁽¹⁾.

Patient self-management of symptoms consistent with mild and moderate exacerbations

Include in the patient's COPD Action Plan guidance on the appropriate self-management of symptoms consistent with mild and moderate exacerbations as per the [Therapeutic Guidelines: Treatment of a COPD exacerbation](#) ⁽⁶⁾:

- Instruct patients to use their bronchodilator as often as needed. Clearly detail the maximum dose that can be taken at one time, and the maximum dose that can be taken in one 24-hour period. Ipratropium is contraindicated in patients taking a LAMA and has a slower onset of effect than salbutamol.
- Provide and dispense a prescription for up to **5 days treatment of prednis(ol)one with no repeats** as part of the patient's written COPD Action Plan to self-initiate if symptoms persist or recur within four hours after reliever use.
- Instruct patients that they must arrange **medical practitioner review as soon as possible after starting prednis(ol)one** for the management of worsening COPD symptoms.
- Antibiotic therapy is only required when an exacerbation is triggered by a bacterial infection ^(6, 28). Refer a patient suspected of having a bacterial infection to an appropriate healthcare provider.
 - Signs suggestive of bacterial infection or pneumonia include increased sputum volume, increased purulence and a possible change in colour, abnormally rapid breathing while at rest, increased heart rate (over 100 bpm) and rigors ⁽²⁸⁾. Note: sputum colour is an unreliable indicator of bacterial infection (the green is caused by immune system proteins binding iron found in the mucus, not bacterial growth directly) and needs to be interpreted in the context of the other signs and symptoms.
 - Sputum culture is not routinely recommended for COPD exacerbations ⁽²⁸⁾.
- Advise all patients to return to the pharmacist (or their chosen healthcare provider) for review of their COPD Action Plan after an exacerbation, even if it was mild and self-managed.

Advise patients to seek emergency care/call an ambulance (even if they have self-initiated prednis(ol)one) if:

- they have severe breathing difficulties
- they have chest pain, 'fluttering' in the chest or an abnormally slow or racing heartbeat (arrhythmia) (COPD exacerbations may present similarly to cardiovascular disease)
- they cannot manage or function at home e.g., can't walk between rooms, eat or sleep due to breathlessness
- they cannot tolerate oral prednis(ol)one
- they have drowsiness or confusion that may indicate hypercapnia or risk of hypercapnia
- they notice the onset or worsening of central cyanosis
- oxygen saturation deteriorates towards 92% SpO₂ ^(29, 30)
- they have leg swelling (peripheral oedema)
- they notice fluctuating blood pressure
- bacterial infection or pneumonia is suspected.
 - symptoms remain uncontrolled despite treatment with an inhaled bronchodilator and prednis(ol)one.

J: Confirm management plan is appropriate

Consult the *Therapeutic Guidelines* ⁽²⁾, *Australian Medicines Handbook* ⁽³⁾ and other relevant references to confirm that management is appropriate, including:

- suitability of inhaler device
- contraindications and precautions
- drug interactions
- lactation status.

K: Communicate agreed management plan

Provide comprehensive advice (including supporting written information when required) to the patient regarding:

- individual product and medicine use (e.g., dosing and inhaler technique)
- how to manage adverse effects
- non-pharmacological measures
- how to self-manage their COPD (see [Develop and implement a management plan](#))
- when to seek further care and/or treatment
- when to return for clinical review.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources provided to patients (and/or caregivers if applicable), and compliance with all copyright conditions.

L: Collaborative care

Provide a copy of the patient's COPD Action Plan to their usual healthcare provider (and any other relevant healthcare professionals involved in the patient's respiratory care). The communication should also include (if appropriate):

- relevant medical history, results of any spirometry testing conducted and level of symptom control (CAAT score)
- changes to existing pharmacotherapy prescribed for COPD or new pharmacotherapy (including for exacerbations)
- a summary of advice provided to the patient including recommendations for multidisciplinary care and referrals
- the next scheduled review appointment.

Ongoing management and monitoring

M: Clinical review and ongoing collaboration

All patients should undergo regular clinical review to participate in the Program, as per the current online version of the [Therapeutic Guidelines](#) ⁽²⁾ and outlined below:

- Review the patient **3 months after management is first commenced and/or new pharmacotherapy is initiated**, including step up of treatment.
- Review the patient after **6 months of stable pharmacotherapy** (with no increase in symptom frequency or severity, and no exacerbations).
- Review the patient as soon as possible (and consider referral if required):
 - if there is deterioration in the condition (increased severity and frequency of symptoms)
 - where symptoms have not improved within a reasonable timeframe following changes to pharmacotherapy
 - if the patient is experiencing intolerable adverse effects
 - after any exacerbation.

At each appointment (scheduled or unscheduled), review:

- the patient's ongoing eligibility for the Program
- patient history to reflect changes in the preceding period
- symptom control using the CAAT and mMRC dyspnoea scale
- inhaler technique and adherence to pharmacological and non-pharmacological treatment including:
 - demonstrate appropriate inhaler technique and reinforce treatment. If a patient is unable to use a device correctly after continued instruction and demonstration, a different device should be tried.
 - inspect equipment to check for breakage or blockage.
 - review the benefits and adverse effects of pharmacotherapy.
- the COPD Action Plan - consider whether the patient's pharmacotherapy is adequate and reinforce non-pharmacological measures
- the development of co-morbidities or factors that may be associated with COPD that may require additional management to improve COPD and health outcomes ([C: Patient history](#)).

Provide an update to the patient's usual healthcare provider following each occasion of care including when any changes are made to the patient's management plan. Proactive, planned and/or unplanned review may also occur with the patient's usual healthcare provider at any time while the patient is enrolled in the Program.

Patients may continue to participate in the Program providing:

- their condition remains eligible to be managed in the Program
- they wish to remain in the Program and continue to consent
- they attend scheduled reviews.

Refer all patients (with their consent) who are no longer eligible or suitable for management within the Program, or who do not wish to continue in the Program, to an appropriate healthcare provider.

Appendices

Appendix 1 - Known risk factors for COPD ^(2, 4)

- tobacco or other types of smoking e.g., hookah
- exposure to second-hand smoke
- prenatal parental smoking
- premature birth
- childhood respiratory illnesses
- infections, particularly tuberculosis
- asthma
- chronic bronchitis
- occupation exposure to dust and fumes e.g., agriculture and livestock farming, plastics manufacturing, mining, food and drinks processing
- genetic susceptibility
- exposure to biomass fuel smoke from cooking or heating (particularly for patients that have come from low- and middle-income countries).

NB: Almost all cases of COPD are the result of an interaction between genetic and environmental factors ⁽⁴⁾.

Appendix 2 – Differentiation between COPD and asthma

In an adult with chronic respiratory symptoms (dyspnoea, cough, chest tightness, wheeze) ^(1, 4, 11, 12, 31)			
	Suggestive of COPD if several of the following features:	Highly suggestive of asthma if several of the following features:	Suggestive of co-existing asthma-COPD if features of both asthma and COPD:
Onset	after age of 40	before age of 40	may have started before or after age of 40
Airflow limitation	- persistent expiratory airflow limitation - with or without bronchodilator reversibility)	- variable expiratory airflow limitation - persistent airflow limitation may be present	- persistent expiratory airflow limitation - with or without bronchodilator reversibility
Symptoms	- persistent dyspnoea (occurs on most days) which limits physical activity and may have been preceded by cough or sputum - are slowly progressive - have limited relief from bronchodilator	- vary over time and in intensity - are associated with certain triggers (e.g. laughter, exercise, allergens) - are worse at night or in the early morning - are recurrent or seasonal - improve with bronchodilators (minutes) or ICS (days to weeks), or may resolve spontaneously	- intermittent or episodic - any of the asthma features in the previous column (e.g., symptoms are associated with certain triggers, symptoms improve with bronchodilators/ICS or spontaneously)
Other	- family history of COPD - history of: - smoking, or exposure to other noxious agents (e.g., smoke/dust) - recurrent chest infections - respiratory illnesses (such as tuberculosis) - low birth weight - no past or current diagnosis of asthma	- personal or family history of allergies (e.g., allergic rhinitis, atopic dermatitis) - family history of asthma - current or childhood asthma diagnosis	- may have a history of: - smoking, or exposure to other noxious agents (e.g., smoke/dust) - respiratory illnesses (such as tuberculosis) - low birth weight - current or childhood asthma diagnosis

Appendix 3 - Classification of COPD severity ^(1, 2, 4)

Severity	Features
Mild	• few symptoms with little to no impact/limitations on daily activities • breathless on moderate exertion • cough and sputum production • FEV₁ is 60–80% of predicted.
Moderate	• breathlessness walking on flat ground • recurrent chest infections • increasing limitations on daily activities and quality of life • moderate exacerbations experienced (oral corticosteroids and/or antibiotics required) • FEV₁ is 40–59% of predicted.
Severe	• breathlessness on minimal exertion • severe limitations on daily activities and impacts on quality of life • increasingly frequent exacerbations and severe exacerbations (hospitalisation required) • FEV₁ is less than 40% of predicted.

Appendix 4 – Interpretation of spirometry ^(4, 15, 16, 32)

The threshold for defining persistent airflow limitation (and when airflow limitation is not fully reversible), is currently accepted to be **FEV₁/forced vital capacity (FVC) ≤0.7, with an FEV₁ of <80% of the predicted value (post-bronchodilator)** ⁽⁴⁾.

- Spirometry should be repeated if FEV₁/FVC results are around the 0.7 threshold.
- Consider asthma (or co-existing asthma and COPD) if there is a large increase in post-bronchodilator FEV₁ (with greater confidence if increase is >400 mL and >15%) (i.e. indicating that airflow limitation is fully or substantially reversible) ⁽⁴⁾. Patient and clinical history should be considered alongside spirometry results. See the *Improved Asthma Symptom Control Program*.
- Indicative flow-volume loop patterns can be found in the [Therapeutic Guidelines: Spirometry](#) ⁽¹⁶⁾, and the [COPD-X Plan: Spirometry](#) ⁽⁴⁾.

NB: The fixed ratio cut-off is arbitrary and may lead to over-diagnosis in older populations, under-diagnosis in younger populations and gender imbalances ⁽⁴⁾.

NB1: The category GLI-2012 'other/mixed' in the GLI 2012 dataset should be used to analyse the spirometry results for Aboriginal and Torres Strait Islander people ^(15, 32).

Resources for interpreting spirometry

- [Therapeutic Guidelines: Pulmonary function testing](#) ⁽¹⁶⁾
- [COPD-X Plan, section C2.3 Spirometry](#) ⁽⁴⁾
- [National Asthma Council Australia Spirometry Handbook for Primary Care](#) ⁽¹⁵⁾
- [European Respiratory Society GLI calculator](#) ⁽³³⁾
- [Global Lung Initiative \(GLI\) 2012 dataset1](#) ⁽³²⁾.

Appendix 5 - Program lifestyle prescription to accompany COPD Action Plan

For an editable version, see [COPD program lifestyle prescription template](#).

Page 1			
Queensland Community Pharmacy Chronic Conditions Management Pilot COPD Monitoring Program Lifestyle Prescription Plan date:			
Patient details			
Name:		Date of birth	
Date of enrolment in the COPD Monitoring Program:			
Patient support person and/or caregiver:	Name, relationship and phone number of someone who assists the patient with their home care		
Program pharmacy details			
Pharmacist name		Phone number	
Pharmacy name and address		Opening hours	
Next COPD Monitoring Program appointment:			
My current FEV ₁ :	Date:	Date of last vaccinations	(list all applicable)
Current level of symptom control (CAAT score):	Date:	Date next vaccinations are due	(list all applicable)
mMRC dyspnoea score:	Date:		
Healthcare team			
General practitioner and clinic:	Name, address and phone number	Closest 24-hour emergency services:	Name, address and phone number
Pulmonary rehabilitation program	Name, address and phone number	Occupational therapist	Name, address and phone number
Dietitian	Name, address and phone number	Physiotherapist /Exercise physiologist	Name, address and phone number
Speech pathologist	Name, address and phone number	Psychologist/ psychiatrist	Name, address and phone number
Other services	Find a Service - Lung Foundation Australia		

Lifestyle prescription

My risk factors and triggers	List identified risk factors and triggers and strategies to minimise and avoid exposure
Exercise plan	- Enter recommendations for physical activity for the patient's age and capability based on the national guidelines: - Informal exercise e.g. building exercise into everyday activities - Formal exercise e.g., walking (moderate intensity) for 30 minutes 5 days of the week strength building 2 days per week - individualised guidance for building up to recommendations - If required: Refer to a GP, exercise physiologist, physiotherapist or other supports for safe exercise
Weight management and diet and nutrition	- Recommendation for weight loss or gain (if applicable) - Summary of nutritional advice for a balanced diet - more of/ increase... - less of/ limit... - If required: Refer to <u>Management for overweight and obesity clinical practice guideline</u> and/or other supports e.g., dietitian
Pulmonary rehabilitation	- Summary of recommendations - Pulmonary rehabilitation program locations
Other management strategies	- If required: Smoking (or vaping) cessation - Refer to <u>Smoking cessation clinical practice guideline</u>, <u>E-cigarette and Vaping Cessation Guide</u> and/or other supports e.g. Quitline or GP - If required: Refer to a GP, psychologist or other clinician for mental health support - If required: Summary of advice regarding alcohol consumption - Non-pharmacological management of breathlessness ^(1, 4) - breathing exercises - breathless recovery position - handheld fans

Document version number	Date	Comments
Version 1.0	01.02.2024	
Version 1.1	11.11.2024	Administrative update.
Version 1.2	01.07.2025	Administrative updates across sections to improve useability of the protocol.
Version 2.0	07.04.2026	Updates across sections to reflect contemporary guidance and improve usability.

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